SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	L.K.PANGAJ	**Reg. No.:**	192511099
Date:	03/08/2026	**Duration:**	60 minutes

Code of Conduct

I (PANGAJ LK /192511099) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: pangaj

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.

1. Identify the Project Vision and Stakeholders

Project Vision

The Employee Cyber Security Awareness and Risk Assessment Portal is designed to improve cybersecurity awareness among employees and reduce organizational security risks through education, assessment, and continuous monitoring. The portal provides interactive learning materials, security awareness quizzes, phishing simulations, and personalized risk assessments. Employees can track their learning progress while administrators monitor security compliance and identify vulnerable users. The system aims to create a strong cybersecurity culture within the organization by promoting secure online behavior and ensuring employees understand current cyber threats. It supports organizations in minimizing human errors, which are among the leading causes of cybersecurity incidents. The portal also generates reports to help management evaluate employee performance and improve training programs. By integrating awareness training with risk assessment, the system ensures that employees not only learn cybersecurity concepts but also apply them in their daily work. The portal is scalable, secure, and suitable for organizations of all sizes.

Project Objectives

- Improve cybersecurity awareness among employees.
- Educate employees about phishing, malware, ransomware, and social engineering attacks.
- Conduct online quizzes and assessments to evaluate employee knowledge.
- Identify employees with high cybersecurity risk levels.
- Generate personalized learning recommendations.
- Monitor employee training completion status.
- Maintain compliance with organizational security policies.
- Provide administrators with analytical dashboards.
- Reduce cyber incidents caused by human error.
- Encourage continuous cybersecurity learning.
- Improve overall organizational security posture.
- Generate automated compliance reports.
- Support remote and on-site employees.
- Ensure secure user authentication and authorization.
- Increase employee confidence in handling cyber threats.

Key Stakeholders

Stakeholder	**Role**
Organization Management	Approves project and reviews security performance
Employees	Complete training, quizzes, and risk assessments
System Administrator	Manages portal users and system configuration
Cyber Security Team	Creates training content and monitors risks
HR Department	Tracks employee training completion
IT Support Team	Maintains technical infrastructure
Compliance Officer	Ensures regulatory compliance
Project Manager	Oversees project execution
Software Developers	Build and maintain the application
Test Engineers	Perform software testing and quality assurance

2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*

Epic 1: User Authentication

User Story 1

As an employee, I want to log into the portal securely so that I can access my cybersecurity training.

User Story 2

As an administrator, I want to manage employee accounts so that authorized users can access the system.

Epic 2: Cyber Security Awareness Training

User Story 3

As an employee, I want to access cybersecurity learning modules so that I can improve my security knowledge.

User Story 4

As an employee, I want to watch educational videos so that I can understand cyber threats visually.

Epic 3: Online Assessment

User Story 5

As an employee, I want to take quizzes so that I can evaluate my cybersecurity knowledge.

User Story 6

As an administrator, I want to view employee assessment scores so that I can identify weak areas.

Epic 4: Risk Assessment

User Story 7

As an employee, I want to receive my cybersecurity risk score so that I know my security awareness level.

User Story 8

As an administrator, I want to identify high-risk employees so that additional training can be assigned.

Epic 5: Reporting Dashboard

User Story 9

As an administrator, I want to generate reports so that management can monitor organizational security awareness.

User Story 10

As management, I want to view graphical dashboards so that I can make informed security decisions.

3. **Define Acceptance Criteria**
    - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

User Story 1 – Secure Login

Given

- The employee has a valid account.

When

- The employee enters the correct username and password.

Then

- The system grants access to the dashboard.

User Story 3 – Access Training Modules

Given

- The employee is logged in.

When

- The employee selects a training module.

Then

- The selected training content should open successfully.

User Story 5 – Take Quiz

Given

- Training has been completed.

When

- The employee starts the quiz.

Then

- Questions should be displayed one by one and the score should be calculated automatically.

User Story 7 – View Risk Score

Given

- The employee has completed assessments.

When

- The employee opens the Risk Assessment page.

Then

- The portal should display the employee's calculated cybersecurity risk score.

User Story 9 – Generate Reports

Given

- Assessment data exists.

When

- The administrator selects Generate Report.

Then

- The portal should generate downloadable reports in PDF or Excel format.

4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

Priority	**Product Backlog Item**	**Business Value**	**Dependency**
High	User Login & Authentication	Essential for system access	None
High	Employee Registration	User management	Authentication
High	Cyber Security Training Modules	Core functionality	Registration
High	Online Quiz System	Employee evaluation	Training
High	Risk Assessment Engine	Risk identification	Quiz
Medium	Dashboard	Performance monitoring	Assessment
Medium	Reports Generation	Compliance	Dashboard
Medium	Notifications	Employee reminders	Dashboard
Low	Discussion Forum	Employee collaboration	Authentication
Low	Feedback Module	User improvement	Training

Prioritization Justification

- Authentication is developed first because every feature depends on secure login.
- Training modules are the primary function of the system.
- Quizzes measure employee learning outcomes.
- Risk assessment identifies vulnerable employees.

- Dashboards and reports assist management in monitoring progress.
- Notifications improve user engagement.
- Discussion forums and feedback modules are useful but not essential for the first release

5. **Plan the Sprint Backlog**
    - Select high-priority user stories for the first sprint and divide them into actionable development tasks.

      Sprint Duration

      2 Weeks

      Sprint Goal

      Develop the core features required for secure employee access and cybersecurity awareness training.

User Story	**Development Tasks**
User Login	Design login page, database connection, authentication, password encryption
Employee Registration	Registration form, validation, database integration
Training Module	Create course pages, upload videos, learning materials
Quiz Module	Develop quiz interface, scoring logic, result display
Dashboard	Design employee dashboard showing progress

Sprint Tasks

- Requirement analysis
- UI/UX design
- Database creation
- Backend API development
- User authentication implementation
- Training content integration
- Quiz development
- Dashboard implementation
- Unit testing
- Integration testing
- Bug fixing
- Sprint review
- Sprint retrospective

6. **Estimate Effort Using Story Points**
    - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

User Story	**Story Points**	**Justification**
User Login	**3**	Simple authentication with validation
Employee Registration	**5**	Form validation and database integration
Training Module	**8**	Uploading videos, documents, and progress tracking

User Story	Story Points	Justification
Quiz System	**8**	Complex scoring and answer validation
Dashboard	**5**	Displays charts and learning statistics
Risk Assessment	**13**	Includes calculations, analytics, and recommendations
Reports	**5**	PDF/Excel report generation

Estimation Technique

The **Fibonacci Story Point Estimation** technique (1, 2, 3, 5, 8, 13, 21) is used because it effectively represents increasing complexity and uncertainty. Story points consider the effort, technical complexity, potential risks, and time required for implementation. Simple features such as login require fewer points, while modules like risk assessment receive higher points due to advanced logic and testing requirements.

7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

     Sprint Goal

     The goal of Sprint 1 is to develop a secure and functional foundation for the Employee Cyber Security Awareness and Risk Assessment Portal. By the end of the sprint, employees should be able to register, log in securely, access cybersecurity learning materials, complete awareness quizzes, and view their training progress. The sprint focuses on delivering the minimum viable product (MVP) while ensuring security, usability, and reliability.

     Expected Sprint Deliverables

- Secure user authentication and login system.
- Employee registration and profile management.
- Cybersecurity awareness training modules.
- Interactive online quiz system.
- Employee dashboard displaying learning progress.
- Database for user information and quiz results.
- Basic administrator dashboard.
- Initial cybersecurity risk assessment framework.
- Unit-tested and integrated application modules.
- Sprint review documentation and product demonstration.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.

Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	**Mark**
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25